

095344

Course Name: Digital Image Processing**Course Outcome**

- CO1: Understand mathematical formulation of an image, its processing steps and relationship between image pixels.
 CO2: Apply Image enhancement using intensity transformations and spatial filtering.
 CO3: Analyze image enhancement for frequency domain using Fourier transform.
 CO4: Formulate region of interest through morphological operations.
 CO5: Evaluate strongly co-related regions obtained through Segmentation using discontinuity and homogeneity based segmentation techniques
 CO6: Describe an object of an image using Shape Number and Boundary descriptors.

Printed Pages: 04**University Roll No.****End Term Examination, Odd Semester 2024-25****B.Tech (CS/IIoT), Year: III, Semester: V****BCSE 0101, Digital Image Processing****Time: 3 Hours****Maximum Marks: 50**

Instruction for students:

1. Use of calculator is allowed
2. All questions are mandatory

Section - A**Attempt All Questions****4 X 5 = 20 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Suppose there is an image I_1 having the 20 rows and 10 columns. There is another image I_2 having 100 rows and have the same aspect as of I_1 . In image I_2 , the maximum value among the all pixels is not exceeding 16. The image I_2 needs to be transferred from one place to other that have a transfer speed of 20 bits/sec. The image is transferred in the form of packet having the packet size of 10 bits and for segregating the packet they are using the flag bits at the beginning and end of every packet such that the bits used in the flag are $1/2$ of the packet size. Compute the transfer time in sec.	4	1	An	C
2	A filter function is given by the sequence $x(n) = \{1, 2, 1\}$. Find the sequence $y(n)$ obtained by two convolution operation as shown below. $y(n) = (x(n) * (x(n) * x(n)))$	4	2	E	M

An image I_3 of 9×9 contains a darker object in a lighter background as shown in the Fig 1.

11	13	10	11	14	11	13	9	11
8	11	13	12	10	11	10	13	12
10	9	3	5	2	1	4	3	10
8	11	2	4	3	2	0	4	11
11	12	3	6	2	3	3	1	12
9	10	1	3	4	4	5	2	11
13	11	2	2	3	4	3	4	10
12	13	3	4	3	2	5	3	12
11	12	11	9	12	11	11	9	10

Fig. 1

- Generate the histogram of the image in tabular form.
- Suggest the gray value threshold to demarcate the object from the background pixels.
- Find the mean of the object pixels.

Suppose an image I_4 is given as shown in the Fig 2.

4	4	4	3	4	4	16	16
4	4	4	3	4	4	18	18
3	3	6	6	4	4	20	20
5	7	6	6	4	4	20	20
8	8	10	10	14	14	14	14
8	8	10	10	14	14	14	14
12	12	12	12	14	14	14	14
12	12	12	12	14	14	14	14

Fig. 2

Apply the region splitting on the image by considering the absolute threshold difference ≤ 4 .

Suppose there is an image I_5 of 7×7 on which a Sobel filter was applied and the result of masking is shown in Fig 3.

	0	0	0	0	0	
	0	0	0	0	0	
	0	-1	-3	-4.1	-4.1	
	0	-1	-3	-4	-4	
	0	0	0	0	0	

$I^o S_H$

	0	4	4.1	0	0	
	0	4	4.1	0	0	
	0	3	3	0	0	
	0	1	1	0	0	
	0	0	0	0	0	

$I^o S_V$

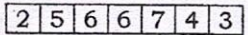
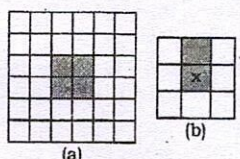
Fig. 3

5	A. Find gradient magnitude and gradient orientation. B. Apply non-maxima suppression as per canny edge detector and show the resultant gradient magnitude image. OR By using Hough transformation by populating an approximate accumulator in parameter space. Assume m has a range (0.8, 1.2) and an interval of 0.1.	4	5	An	C
	 Fig. 4 Find the slope m and offset c of the digital line ($s = mp+c$) provided as shaded pixels in the Fig 4.				

Section - B

Attempt All Questions

3 X 5 = 15 Marks

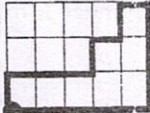
No.	Detail of Question	Marks	CO	BL	KL
6	Consider an image as shown in Fig. 5 (alias I_6). Suppose we apply a 1×3 average filter on an image I_6 to get the image I_7. Again, we apply the same filter on image I_7 to get the image I_8.  Fig. 5 Find the final value of the center pixel.	3	2	C	P
7	Given a binary image I_9 as shown in Fig 6(a) and a structure element S as shown in Fig. 6(b) in which shaded area represent the object pixels and x is the origin.  Fig. 6 Find the dilated image using the formula $I \oplus S = \{p p = i + s, i \in I, s \in S\}$	3	4	An	C
8	Discuss the steps of the global thresholding.	3	5	R	F
9	Suggest the suitable sensor resolution for the following scenario.	3	1	C	C

	A. A user would like to monitor the growth of wheat crop from the sowing time till it gets harvested. B. A tropical cyclone arises in the southern part of Bay of Bengal and is moving upwards which is tracked by a user. C. A user would like to have a sensor where he is not bothering about the pixel size but he wants more number of shades in the image.				
10	How does 1st order derivative help in edge detection? Where can we use 2nd order derivative in edge detection?	3	3	U	C

Section - C

Attempt **All** Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL																																																															
11	Compute the Fourier domain sequence $X(k)$ for the time domain sequence $x(n)$ by applying DFT. $x(n) = \{1, 1, 0, 0\}$	5	3	E	M																																																															
12	Considering 4 direction chain code, dot is the starting point and move in clockwise direction.  Fig. 7 Find the shape number of the highlighted shape shown in Fig. 7.	5	6	E	P																																																															
13	Use the region growing approach to segment the object in the image as shown in Fig. 8. The seed pixel in the object is highlighted. Region will grow when the difference between the two pixel value is less than 5. Segment the final object and count the total number of pixels in the object. <table><tr><td>11</td><td>23</td><td>22</td><td>25</td><td>20</td><td>22</td><td>26</td><td>19</td><td>22</td></tr><tr><td>14</td><td>25</td><td>16</td><td>14</td><td>11</td><td>23</td><td>16</td><td>21</td><td>18</td></tr><tr><td>25</td><td>20</td><td>27</td><td>26</td><td>18</td><td>10</td><td>13</td><td>23</td><td>15</td></tr><tr><td>26</td><td>18</td><td>13</td><td>13</td><td>22</td><td>24</td><td>17</td><td>27</td><td>13</td></tr><tr><td>10</td><td>22</td><td>12</td><td>11</td><td>28</td><td>13</td><td>21</td><td>12</td><td>24</td></tr><tr><td>17</td><td>11</td><td>25</td><td>21</td><td>17</td><td>28</td><td>28</td><td>18</td><td>26</td></tr><tr><td>24</td><td>15</td><td>11</td><td>12</td><td>28</td><td>29</td><td>29</td><td>24</td><td>25</td></tr></table> Fig. 8	11	23	22	25	20	22	26	19	22	14	25	16	14	11	23	16	21	18	25	20	27	26	18	10	13	23	15	26	18	13	13	22	24	17	27	13	10	22	12	11	28	13	21	12	24	17	11	25	21	17	28	28	18	26	24	15	11	12	28	29	29	24	25	5	5	An	M
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